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ORIGINAL ARTICLE NON-SURGICAL PROCEDURES

The Long-Term Efficacy of Botulinum Toxin Type A for Lower Facial Rejuvenation in 400 Patients

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Abstract

Background Botulinum toxin type A (BoNT-A) has become an established treatment for upper facial rejuvenation; however, its use in the lower face and neck is less explored and presents anatomical challenges. This study evaluates the long-term efficacy and safety of BoNT-A injections for lower facial rejuvenation, emphasizing potential cumulative effects and muscle modulation.

Methods A retrospective single-centre analysis was conducted on 400 patients (380 females and 20 males; aged 35–85 years) treated between January 2018 and December 2024. Patients received, during a full face treatment, BoNT-A injections targeting the mentalis, depressor anguli oris, platysmal bands, and masseter muscles, with standardized doses ranging from 120 to 160 units per session every 6 months. Patients underwent a minimum of three sessions over a follow-up period of 24–60 months. Outcomes were assessed using the Merz Aesthetic Scale (for platysmal bands) and a 5-point Visual Analogue Scale (VAS) based on standardized pre-treatments photographic and video documentation reviewed by two independent plastic surgeons.

Results The mean Merz score reported these results baseline (pre-first treatment): 4.2 ± 0.6, after first cycle: 2.1 ± 0.4 ($p < 0.01$), and after third cycle: 1.8 ± 0.3 ($p < 0.001$ compared to baseline). Independent VAS ratings confirmed significant improvements in contour and soft tissue balance Global improvement score: pre-treatment 1.5 ± 0.7 → post-treatment 4.1 ± 0.4 ($p < 0.001$). Notably, patients maintained results longer than the expected pharmacological effect window, supporting the hypothesis of functional muscle retraining. Mild adverse events occurred in less than 12% of patients, mainly localized bruising and discomfort, with no serious complications.

Conclusions In this large cohort of 400 patients, BoNT-A injections for lower facial rejuvenation produced significant aesthetic improvements with progressive and sustained benefits over time. The concept of cumulative neuromodulation offers new insight into long-term lower face aesthetic management. Our results support BoNT-A as a safe, effective, and minimally invasive treatment option in expert hands.

Level of Evidence II This journal requires that authors 43 assign a level of evidence to each article. For a full 44 description of these Evidence-Based Medicine ratings, 45 please refer to the Table of Contents or the online 46 Instructions to Authors http://www.springer.com/00266.

Keywords BoNT-A · Botox · Full face · Cosmetic medicine

Introduction

Botulinum toxin type A (BoNT-A) has revolutionized aesthetic medicine since its first cosmetic use in the late 1980s. The product acts by inhibiting the release of acetylcholine at the neuromuscular junction, temporarily preventing muscle contraction, and thus improving dynamic wrinkles and folds [1–3]. Although initially approved and widely applied for the upper face (glabella, forehead lines, and crow's feet), there has been increasing

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interest and clinical use in the lower third of the face and neck [4–6]. These areas are particularly challenging due to the complex anatomy and the powerful interplay of depressor muscles which contribute to the aged appearance of the lower face and cervical contour [7, 8]. The platysma muscle, depressor anguli oris, depressor labii inferioris, mentalis, and masseter muscles play central roles in facial dynamics and aesthetics [9, 10]. The use of BoNT-A in these areas requires expertise to prevent side effects such as asymmetry, speech alterations, or dysphagia [11, 12]. Nevertheless, experienced practitioners have reported excellent outcomes, with improvements in jawline contour, platysmal banding, marionette lines, and overall facial harmony [13, 14]. Emerging data suggests that repeated injections not only maintain clinical effects but may also induce long-term remodelling of neuromuscular interactions through synaptic plasticity [15, 16]. This phenomenon has been termed "functional muscle retraining", whereby chronic hyperactivity of depressor muscles is gradually balanced, leading to more stable aesthetic results even after the temporary pharmacological effect of BoNT-A has waned [17–19]. The aim of this monocentric retrospective study is to evaluate the long-term efficacy and safety of BoNT-A for rejuvenation of the lower face and neck in a large cohort of patients, as part of a full face treatment, analysing both patient-reported outcomes and objective clinical assessments across repeated sessions.

Materials and Methods

Study Design

This retrospective study was conducted in an independent aesthetic medicine centre between January 2018 and December 2024. All patients underwent treatment on the face, but in this study, we focused only on the results of the lower third of the face. Approval from the institutional review board was obtained, and the study followed the Declaration of Helsinki principles. All participants provided informed consent prior to treatment (Tables 1, 2).

Participants

A total of 400 patients were enrolled (380 females and 20 males), aged between 35 and 85 years. Inclusion criteria were:

- Age $\geq$ 35 years
- Request for aesthetic improvement of the lower face and neck
- Minimum of three BoNT-A treatments performed at regular intervals of approximately 6 months

Exclusion Criteria

- Patients receiving concurrent aesthetic treatments during the observation period (dermal fillers, surgery, and laser therapy)
- Neuromuscular disorders (e.g. myasthenia gravis and ALS)
- Pregnancy or breastfeeding
- Pharmacological resistance to the effect of botulinum toxin
- Active skin infections or autoimmune diseases involving the face or neck

Table 2 Sample data

Sample data	Number (*N*=400)
Age	
Mean (years old)	49,5
Gender	
Male	20
Female	380
Follow up	
Mean (years)	2

Table 1 PICOS criteria

Component	Description
Population	400 patients (380 females, 20 males), aged 35-85
Intervention	BoNT-A injections (120-160 U) every 6 months in standardized lower face/neck sites
Comparison	Baseline scores prior to first treatment
Outcomes	Merz Scale, VAS score, photo/video documentation
Study design	Retrospective, single-center cohort study

Injection Protocol

All procedures were performed by two experienced aesthetic doctor. Patients received on average 120–160 units of BoNT-A per session (onabotulinumtoxinA) as a full face treatment, in particular injected in the lower face at predefined anatomical sites (the focus of this study is the lower face and neck treatment):

- Mentalis: 2–6 units
- Depressor anguli oris: 2–3 units per side
- Platysmal bands: 20 per side and 20 for anterior bands
- Masseter (in selected cases): 20–25 units per side

In Figs. 1 and 2, we have highlighted the changes that can be obtained between the middle third of the face and the lower third of the face in patients who have only undergone botulinum toxin infiltration. Figures 1 and 2A represent the pre-treatment and Figs. 1, 2B represent the post-treatment. We have also schematized on the photos the key points of the changes obtained with the treatment. Figures 1 and 2B are 2 months post-treatment.

A systematic protocol was used to ensure consistency. Injections were delivered using 32G needles, and mapping followed published anatomical landmarks to minimize risk [20–22].

Outcome Measures

The primary outcome was improvement in platysmal banding, assessed by the Merz Aesthetic Scale (5-point scale). The secondary outcome was overall aesthetic improvement evaluated via standardized pre-treatments photo and video documentation, scored independently by two blinded doctors using a 5-point Visual Analogue Scale (VAS). Photographs were acquired with a Nikon DSLR camera under standardized lighting and head positioning. Adverse events were documented systematically at each visit.

Statistical Analysis

Descriptive statistics were performed. Paired Student's *t*-test compared pre- and post-treatment scores. A *p*-value of < 0.01 was considered statistically significant. Statistical analysis was conducted using SPSS version 26 (IBM, USA).

Results

All patients completed at least three treatment cycles with a mean follow-up of 36 months (range 24–60 months). No patients were lost to follow-up.

The Merz Aesthetic Scale (Fig. 3) Showed Significant Improvement

- Baseline (pre-first treatment): 4.2 ± 0.6
- After first cycle: 2.1 ± 0.4 ($p < 0.01$)
- After third cycle: 1.8 ± 0.3 ($p < 0.001$ compared to baseline)

The VAS scores (Fig. 4) from independent reviewers confirmed aesthetic improvements:

Global improvement score: pre-treatment $1.5 \pm 0.7 \rightarrow$ post-treatment 4.1 ± 0.4 ($p < 0.001$).

Notably, the progressive improvement observed between sessions suggested that the clinical effect extended

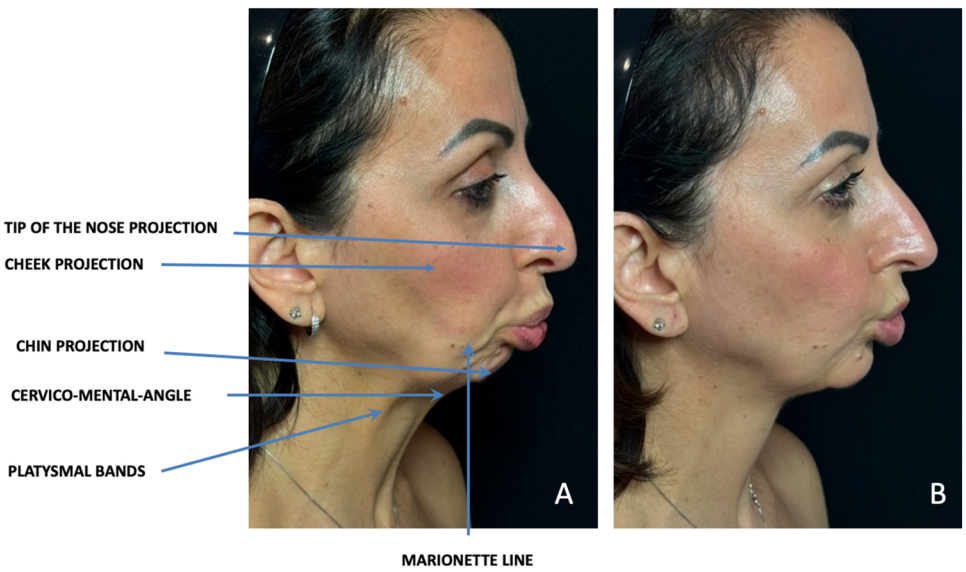


Fig. 1 Diagram of the changes that can be obtained in the middle and lower third of the face. **A** represents the pre-treatment, and **B** is 2 months post-treatment

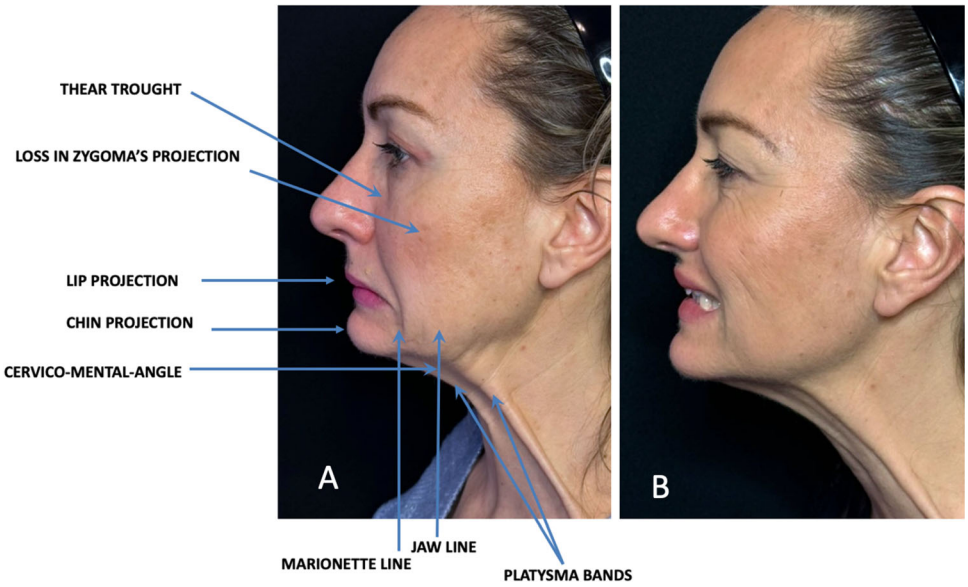


Fig. 2 Diagram of the changes that can be obtained in the middle and lower third of the face **A** represents the pre-treatment, and **B** is 2 months post-treatment

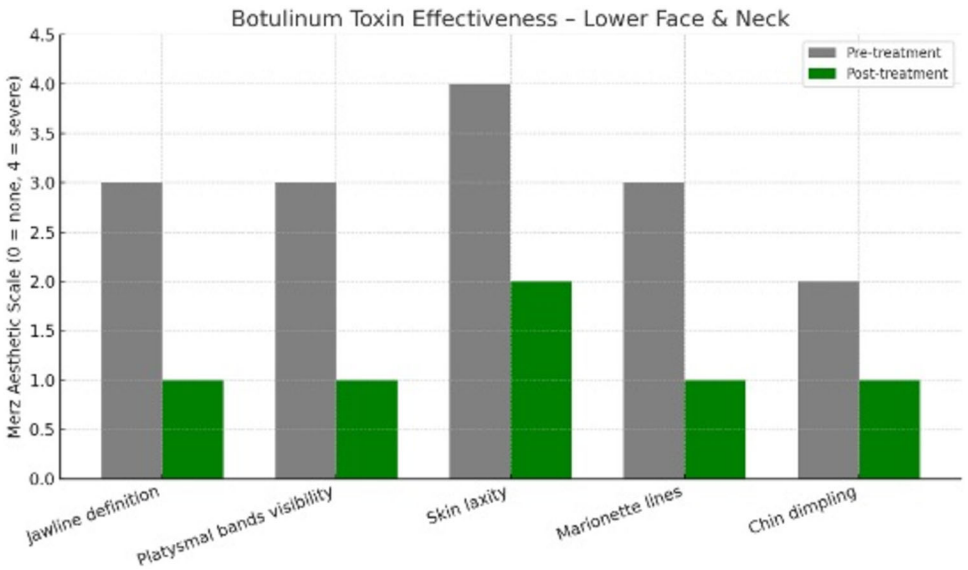


Fig. 3 Results VAS scale

beyond the expected pharmacological duration of 3–4 months.

Adverse Events

- Mild injection site bruising in 12%
- Temporary local discomfort in 8%
- No severe or permanent complications occurred.
- No cases of dysphagia, ptosis, or asymmetry were reported.

In Figs. 5, 6, 7, 8, and 9 we want to show the pre-treatments over the years and how in the absence of botulinum toxin it is still possible to have benefits in the balance of muscle contraction and tissue repositioning, thanks to the frequency of execution of the treatment.

Discussion

The data from our large, monocentric study confirm the significant and progressive aesthetic improvement of the lower face and neck with the repeated use of botulinum toxin type A (BoNT-A) injections over a follow-up period of 2–5 years. These results add valuable evidence to a relatively underexplored area, as the majority of BoNT-A literature focuses on the upper face [1–4]. The rejuvenation of the lower face is notoriously challenging, due to the complex anatomical interplay of depressor muscles such as depressor *anguli oris*, *depressor labii inferioris*, *mentalis*, and *platysma*, which contribute to the descent of facial tissues and age-related disharmony [5–8]. Historically, surgical lifting techniques were considered the only

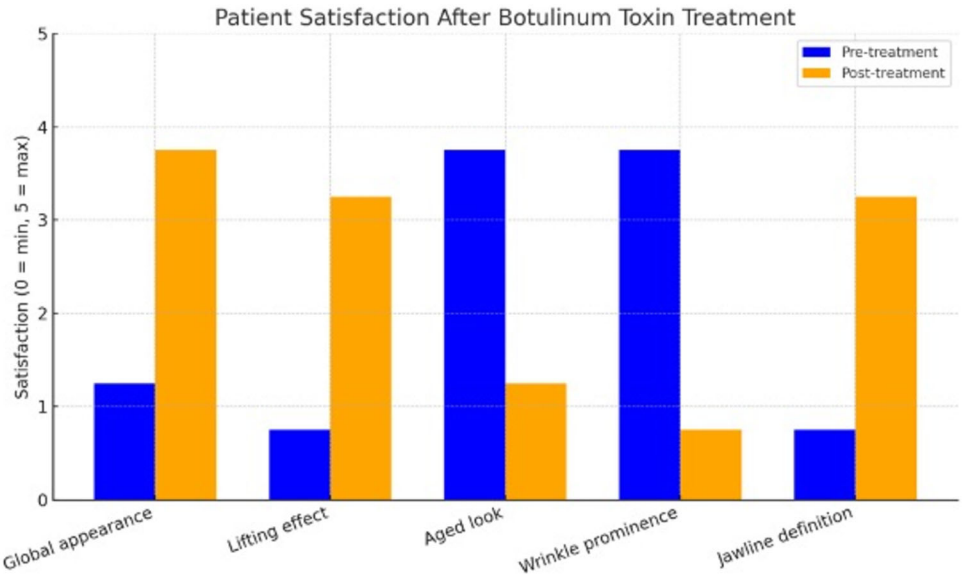


Fig. 4 Results Merz scale



Fig. 5 A 69-year-old patient, pre-treatment from 2020 to 2024

effective long-term solution for this region. However, our findings suggest that BoNT-A may offer a minimally invasive yet powerful alternative or adjunct by rebalancing the depressor-elevator dynamics [9–12]. A remarkable finding of this study is the cumulative and prolonged improvement of outcomes beyond the pharmacological duration of BoNT-A, which classically ranges between 12 and 16 weeks [13, 14]. This supports the emerging hypothesis of muscle retraining and neuromuscular plasticity. It has been proposed that chronic denervation from repeated toxin exposure induces a new muscular balance, reduces muscle hyperactivity, and modifies central motor pattern generation [15–18]. Preclinical animal models have confirmed such remodelling, and our clinical data now support this phenomenon in a human cohort [19–22]. Additionally, BoNT-A has been shown to promote



Fig. 6 A 65-year-old patient, pre-treatment from 2019 to 2024



Fig. 7 A 65-year-old patient, pre-treatment from 2019 to 2024

fibroblast modulation and extracellular matrix remodelling, potentially contributing to improved skin texture and quality in the treated area [6, 23]. Some authors have even hypothesized a role for BoNT-A in modulating local microcirculation and sebaceous gland activity [7, 8]. Although these secondary mechanisms were not directly assessed in our study, they may help explain why patients reported improvements not only in contour but also in overall skin appearance. From a practical standpoint, our study reinforces the importance of standardized injection protocols, precise dosage, and accurate patient selection to minimize risks [9–12]. Our adverse event rate was extremely low, with no severe or permanent complications. This finding is consistent with the literature when injections are performed by experienced injectors using an anatomical approach [13–15]. The reproducibility of the results over a large sample size ($n = 400$) strengthens the external validity of our findings, providing important data to guide clinical practice. Our experience suggests that a minimum of three sessions over 12–18 months is needed to observe



Fig. 8 A 71-year-old patient, pre-treatment from 2019 to 2024



Fig. 9 A 71-year-old patient, pre-treatment from 2019 to 2024

the long-term stabilizing effect. Nonetheless, the study has limitations. First, it is retrospective and non-randomized, which can introduce selection and observer bias. Future prospective, randomized controlled trials should further validate the concept of functional neuromodulation induced by BoNT-A in the lower face [16, 17]. Second, no control group was included for ethical reasons, as it was not possible to leave patients untreated for extended periods once treatment had been initiated and found beneficial. Third, while the photographic and VAS assessments were independently scored, more objective 3D imaging or digital morphometry could improve accuracy in future studies. Another limitation is the demographic homogeneity of the study population, which was predominantly female (95%). Although this reflects real-world aesthetic patient populations, future research should investigate whether gender differences may influence response patterns. Finally, this study focused exclusively on BoNT-A. Comparative

studies evaluating BoNT-A versus combination therapies (e.g. fillers, energy-based devices, or surgical approaches) would be highly valuable to position BoNT-A within an integrated rejuvenation algorithm for the lower face. In conclusion, our findings indicate that BoNT-A, when used following an anatomical and structured approach, offers not only temporary wrinkle relaxation but also a long-term rebalancing of muscle dynamics. This effect can reduce the need for more invasive procedures or act as a preparatory treatment to optimize surgical outcomes. The concept of "chemical lift" in the lower face, often regarded as elusive, seems to be confirmed by this large-scale clinical experience. The principal limitations of the study are: It is a single-centre study, we do not have advanced three-dimensional analysis systems that highlight the improvement but only standardized pre- and post-photos, it is a retrospective study.

Clinical Recommendations

Based on our findings, we suggest

- Patient selection: Careful exclusion of patients with contraindications (neuromuscular disease, infection, and pregnancy).
- Standardization: Strict injection protocols and anatomical mapping should be followed.
- Dosage: Recommend 120–160 units per session, adapted to patient needs.
- Timing: Optimal maintenance observed with treatments at 6-month intervals.
- Safety monitoring: Adverse events were rare and mild, confirming BoNT-A as safe when used correctly.

The data also suggest that repeated lower face treatment may serve as a valuable pre-conditioning step prior to surgical lifting procedures by rebalancing soft tissue tension.

Conclusion

In this large cohort of 400 patients, BoNT-A injections for lower facial rejuvenation produced significant aesthetic improvements with progressive and sustained benefits over time. The concept of cumulative neuromodulation offers new insight into long-term lower face aesthetic management. Our results support BoNT-A as a safe, effective, and minimally invasive treatment option in expert hands.

(A) The authors declare that they have no conflicts of interest to disclose.

(B) This article does not contain any studies with human participants or animals performed by any of the authors.

(C) For this type of study, informed consent is not required.

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